



MECHANISM OF INJURY IN A HIGH ANKLE SPRAIN: A CADAVER STUDY

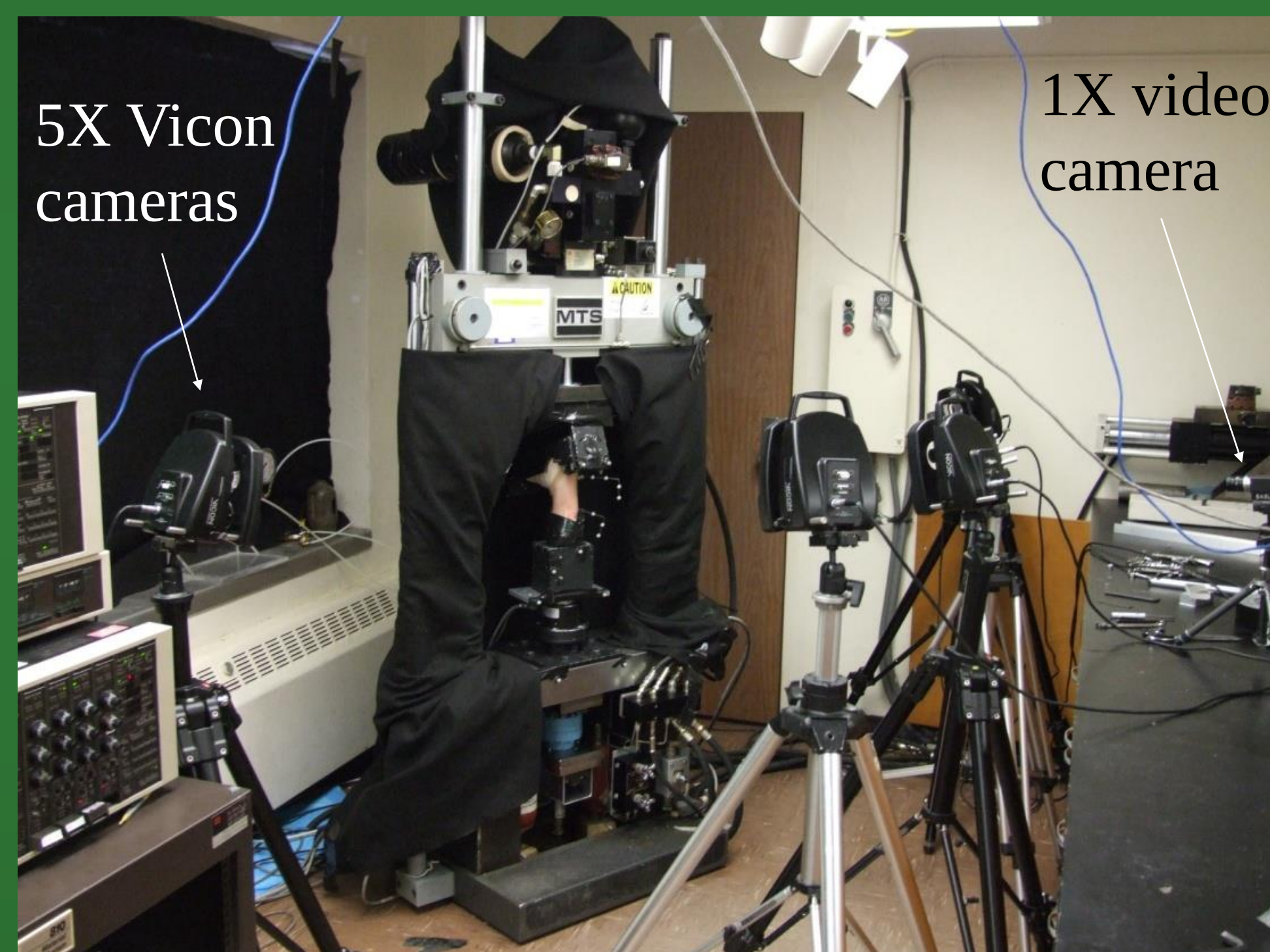
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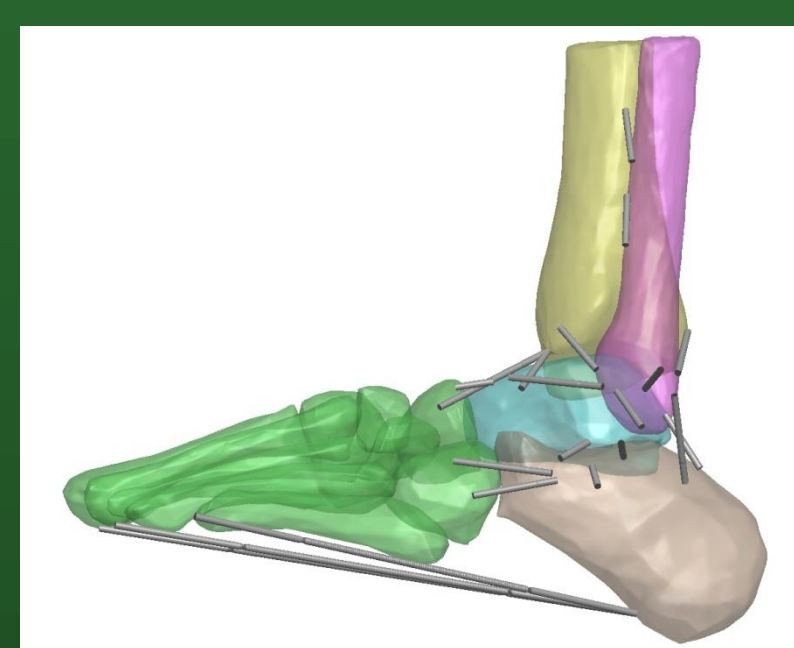
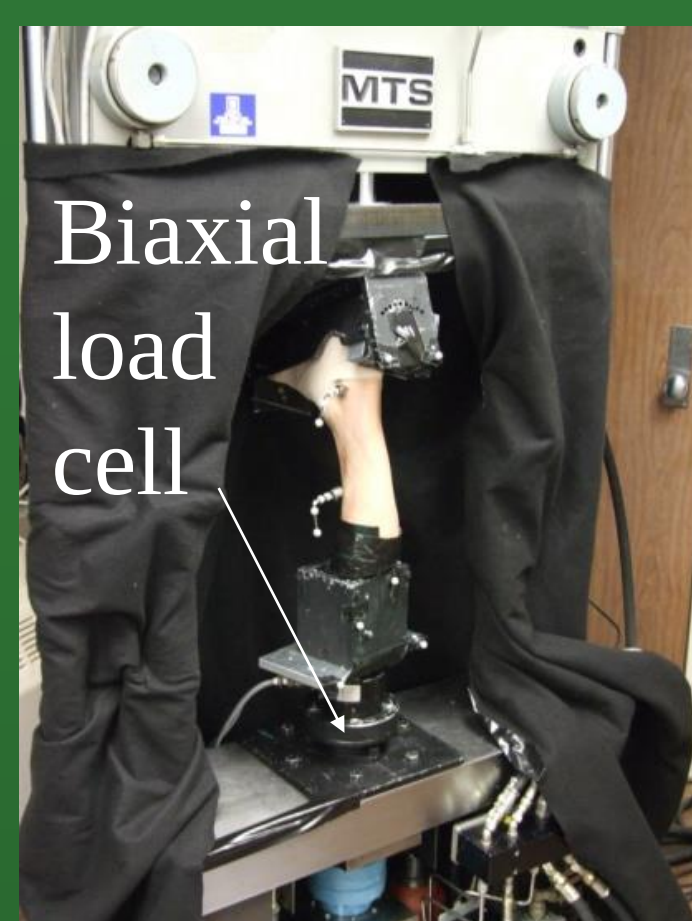
INTRODUCTION

- High ankle sprains are not very common, comprising 1 to 11% of all ankle sprains [4].
- More disabling problem [7]; often misdiagnosed [1].
- Longer recovery period: acute (0-6wks), subacute (6wks–3mos), chronic (>3mos) [1,2].
- In case without widening, surgeons do not have a direct diagnostic test [1].
- Require different treatment [4]; cause chronic ankle dysfunction [5].
- **Mechanism of injury:** classically involve external rotation, but no experimental proof.
- Study by Dattani et al. [3] shows the sequence of injuries due to external foot rotation is deltoid ligament, ATiFL, PTiFL, IOL, and finally a spiral fracture of the fibula.
- Previous studies from this lab indicate the role of eversion in this injury.
- **Hypothesis:** external rotation of a highly everted foot will generate a high ankle sprain (damage to the ATiFL).

METHODS



➤ Taping technique.



➤ Ankle model from Wei et al. [6].
➤ 14 bones and 20 simulated ligaments.

- 6 pairs of cadaver (male) lower extremity.
- Right limbs → neutral in coronal plane.
- Left limbs → highly everted (20 deg).
- Both limbs → taped; 20 deg dorsiflexion.
- Marker sets screwed into tibia and talus.
- External rotation of the foot (internal rotation of the tibia) until gross injury.
- A computational model being used to simulate the testing.

RESULTS

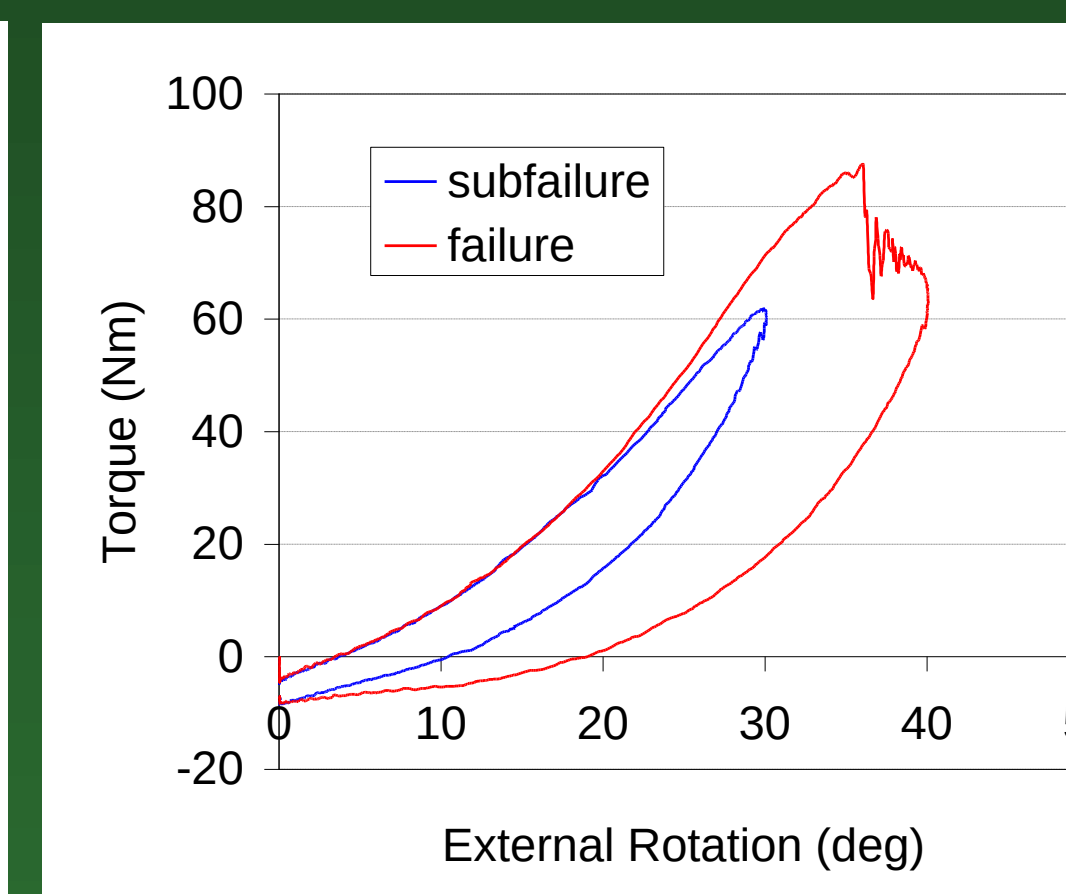
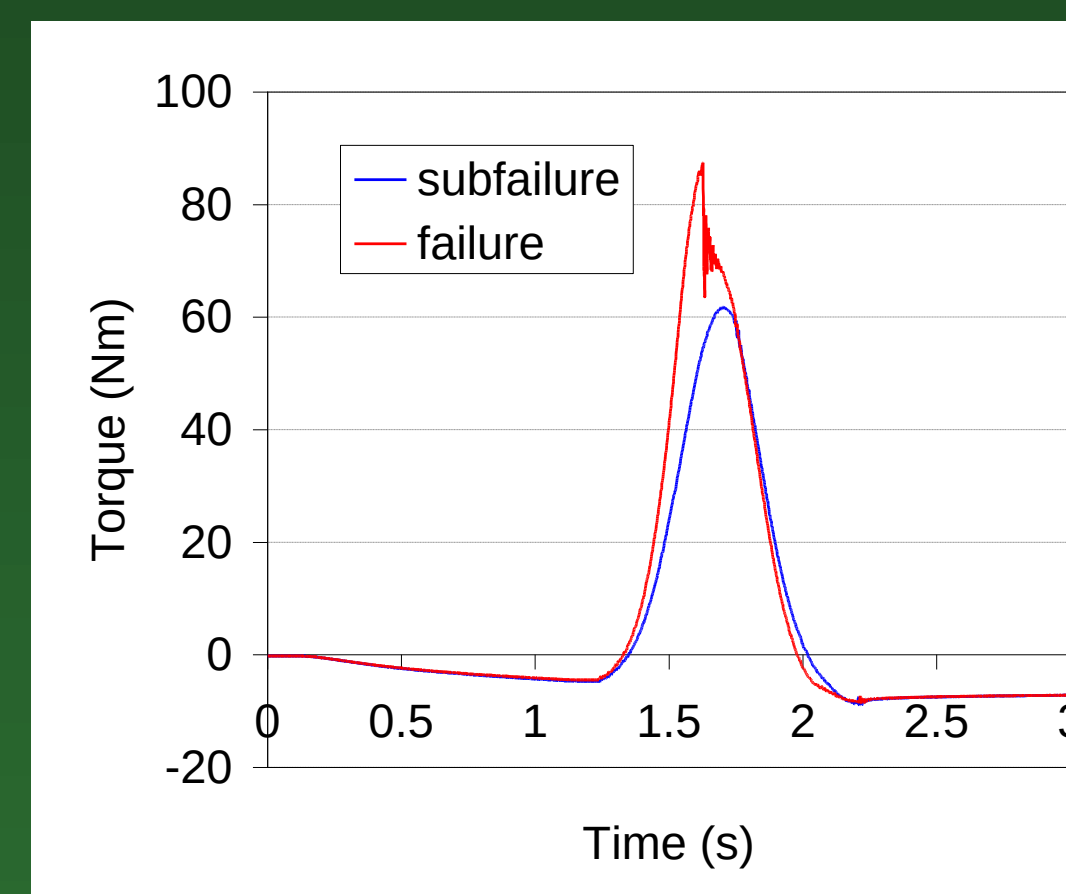


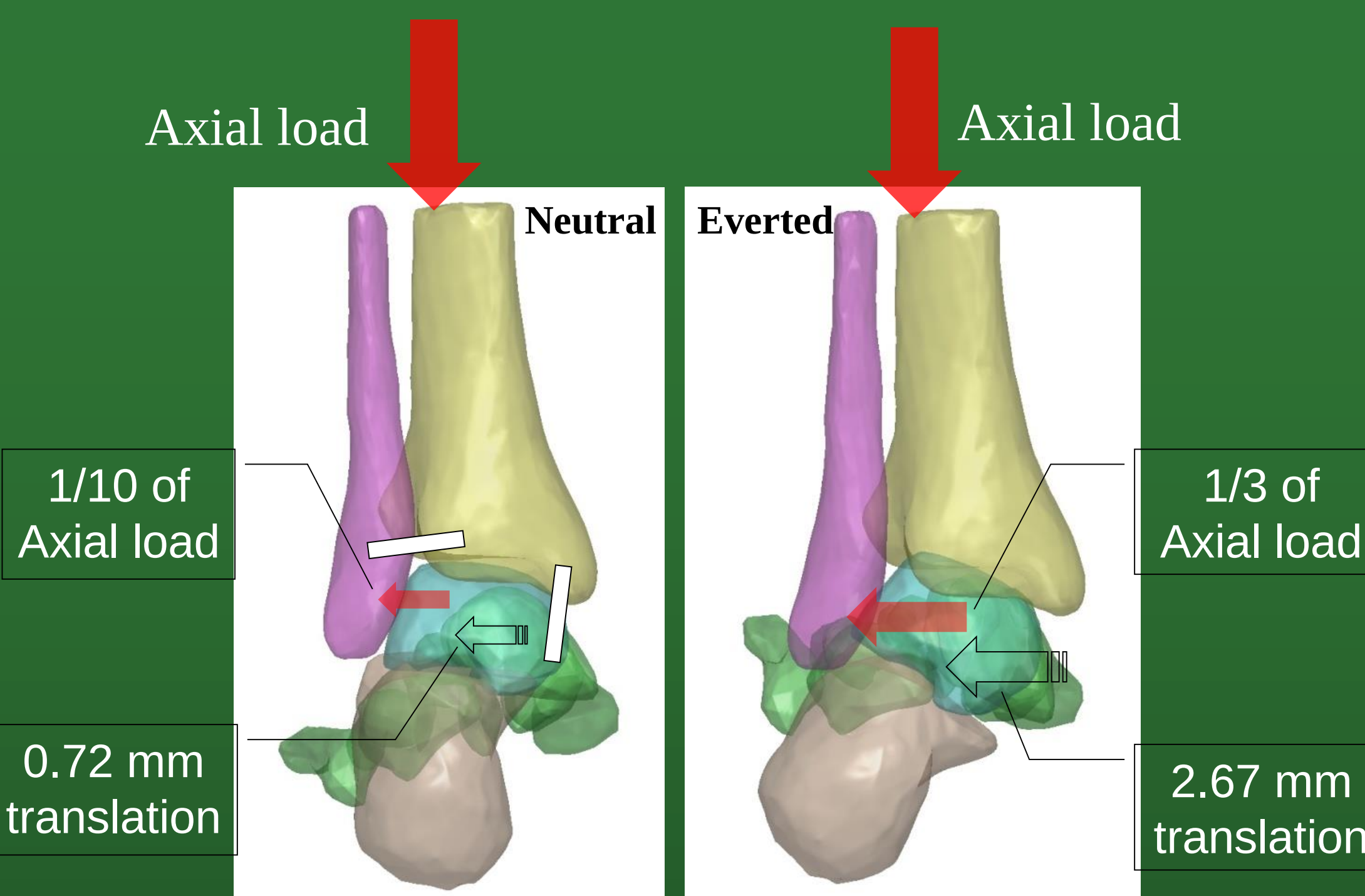
Table 2 Bony contact force and ligament pre-strain generated by an axial load of 1500 N in the computational model.

	Neutral	Everted
Contact force (N) between the talus and the fibula	131	482
Pre-strain (%) in the anterior deltoid ligament	8.5	9.6
Pre-strain (%) in the ATiFL	4.3	14.0

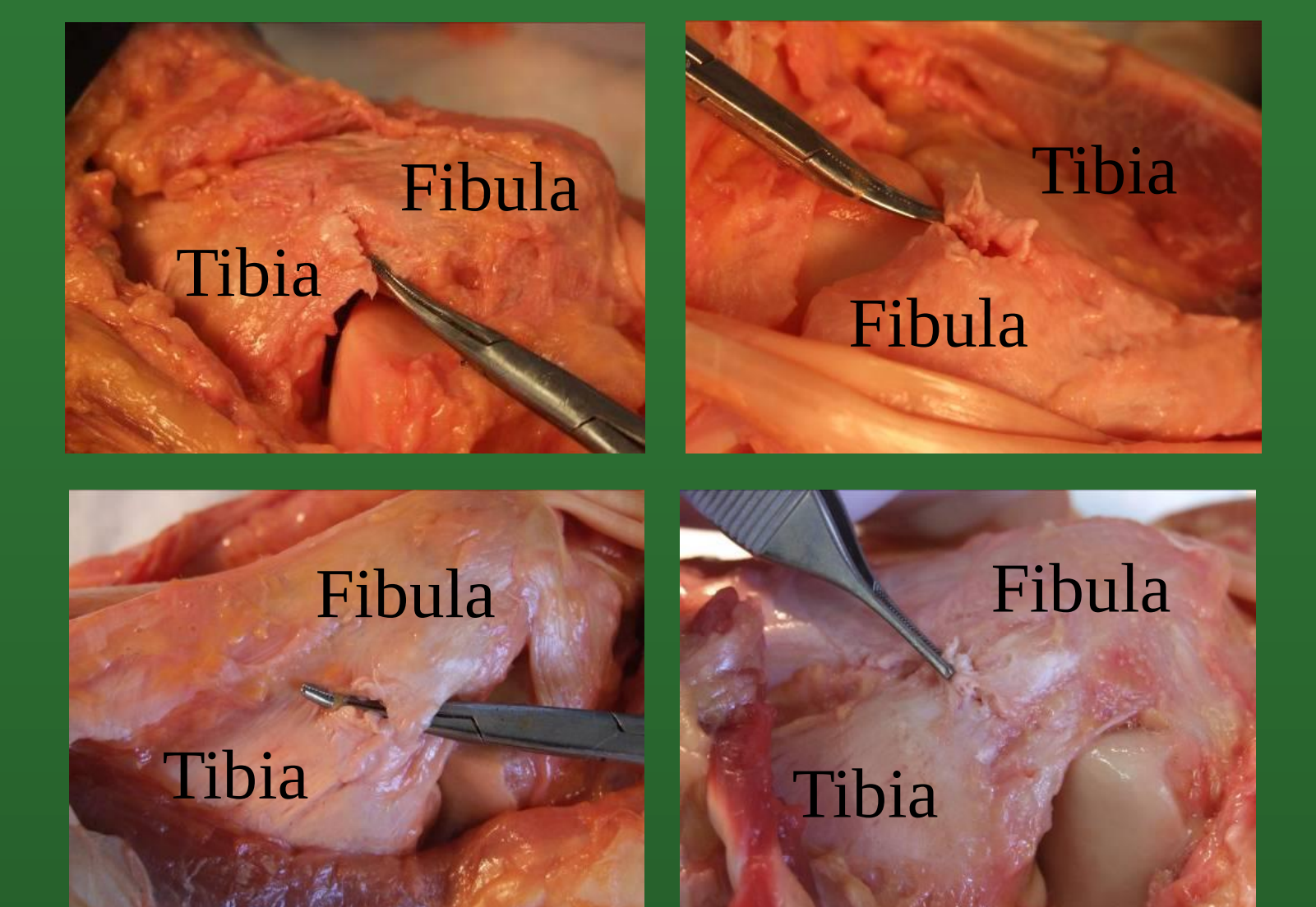
- Typical failure and subfailure curves.
- Large energy lost due to failure.
- Simulation results explanation.

Table 1 Specimen descriptions, torque and foot rotation at failure, and resultant pathology.

Specimen	Age	Height (m)	Weight (kg)	Failure torque (Nm)		Failure rotation (deg)*		Failure mode	
				Neutral	Everted	Neutral	Everted	Neutral	Everted
1	76	1.72	64	87.4	78.3	36.0	43.4	Tibial avulsion of the anterior deltoid	Rupture of the ATiFL
2	56	1.83	79	80.9	91.8	47.1	55.3	Partial tear of the anterior deltoid	Partial tear of the ATiFL
3	40	1.88	75	88.0	94.5	35.1	50.1	Partial tear of the anterior deltoid	Rupture of the ATiFL
4	50	1.75	88	101.5	103.8	39.8	50.0	Rupture of the deltoid	Rupture of the ATiFL
5	55	1.78	66	53.1	87.2	31.1	43.4	Spiral fracture of the fibula	Partial tear of the ATiFL
6	56	1.88	93	95.3	86.5	37.0	38.6	Rupture of the anterior deltoid	Rupture of the ATiFL
Mean	55.5	1.81	77.5	84.4	90.3	37.7	46.8		
SD	11.8	0.07	11.6	16.9	8.6	5.4	6.1		



- Orientation of ATiFL is medial-lateral in coronal plane → direct stretch.
- Orientation of deltoid ligament is antero/postero-distal in sagittal plane → first wrap, and then stretch.



➤ Typical images of rupture (top two) and partial tear (bottom two) of the ATiFL.

DISCUSSION

- Isolated ligamentous injury to the ATiFL due to external rotation of a highly everted foot.
- A highly everted foot may predispose the ATiFL to injury.
- Computational model helped explain the mechanisms.
- Talus motion during axial loading is critical: lateral translation; push the fibula; stretch the ATiFL.
- Avoiding foot eversion may prevent high ankle sprains.

REFERENCES

- [1] Amendola, *Metcalf Memorial Meeting*, 2008; [2] Boytim et al., *Am J Sports Med* 19, 1991; [3] Dattani et al., *J Bone Joint Surg Br* 90, 2008; [4] Hopkinson et al., *Foot & Ankle* 10, 1990; [5] Taylor et al., *Am J Sports Med* 20, 1992; [6] Wei et al., *Ann Biomed Eng* 39, 2011; [7] Wolfe et al., *Am Fam Physician* 63, 2001.